

SignQuest: An American Sign Language Learning Web Application

Group 4: Adam Kolodziejczak, Caleb Radbourne, Ethan Gonsalves, and Om Sathe

Department of Computer Science, Ontario Tech University

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Dr. Michael Miljanovic

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Abstract

SignQuest is an educational web application that is designed to teach American sign language (ASL) through several learning objects such as structured dynamic lessons, several games, dictionary and a live translator. We aimed at creating an application for ASL that addresses the gaps of language learning technologies while also covering universal design for learning principles; engagement, representation, and finally action and expression. Engagement is supported through gamified progression. Representation through multiple learning objectives and modular content. Action and expression through the translator model and other features that provide immediate feedback based on the users actions. After our research SignQues focuses on improving linguistic and metacognitive ability for long term retention and attitudes towards learning, especially language applications that involve repetition. Finally SignQuest is built on two servers, a frontend and backend, where the frontend server uses Next.js and Azure while the backend uses FastAPI a Azure container Apps. The novel feature of this application includes YOLOv11 for live sign rection and games that involve cognitive skills. SignQuest utlamly focus on ASL education outside of a trationdal classroom at a users ownpace and own practice options.

SignQuest: An American Sign Language Learning Web Application

Introduction

Accessibility through communication remains an issue, especially for those who cannot hear well or are deaf, in environments where others do not know American Sign Language (ASL). It is only in such environments that one is limited to writing messages or using simple signs that are neither fast nor expressive enough. This is because ASL itself is a sign language whose meanings are expressed using handshapes, movement, location, and facial expression.

Learning ASL requires the development of both cognitive understanding and motor coordination. Unlike many spoken languages, which can be reinforced through audio- and text-based repetition, ASL relies on visual interpretation, the physical formation of signs, and movement. This creates a barrier for beginners, who must not only memorize vocabulary but also learn precise motion patterns and timing. Without consistent feedback or structured practice environments, learners often struggle to progress after recognition of signs into accurate formation and use in context.

Recent developments in educational technology have shown that interactive systems can support more effective independent learning. Specifically, digital learning environments that incorporate immediate feedback and structured progression have been shown to improve retention and learner engagement. However, many existing language learning tools are optimized for spoken or written languages and do not support learning that relies on movement and visual communication.

This report analyzes the design and implementation of a web based ASL system learning that explores how computer vision techniques can support real time recognition and feedback during practice. It also discusses how structured digital learning environments can be used to reinforce repetition-based learning for a visual language. The remainder of this report covers the educational background, system design decisions and technical implementation of the application.

The Problem and Motivation

American Sign Language (ASL) presents a distinct learning challenge because it relies on hand movements, position, and facial expressions rather than spoken or written language. For beginners, this poses a combined challenge: understanding the meaning of the signs while developing the motor skills needed to reproduce them accurately. In practice, many learners can recognize these signs in isolation but struggle to perform them correctly in real-world contexts, particularly without structured feedback.

The existing learning resources emphasize passive exposure through videos or static references such as dictionaries. While they are useful for memorization, these approaches provide limited opportunities for iterative practice or performance correction. As a result, learners stop progressing after the initial vocabulary stage and lack confidence when it comes time to attempt real communication.

This gap highlights the need for a more interactive learning approach that supports repeated practice, immediate feedback and gradual skill progression. The development of SignQuest addresses this need by exploring how an interactive, technology-supported environment can facilitate the transition from no prior knowledge to sign recognition in real-world contexts.

The Solution

SignQuest is a web-based ASL learning platform that combines structured ASL lessons with interactive practice tools. The system integrates lessons, practice activities, and game-based learning modules into a single system, designed to support incremental skill development.

Rather than relying solely on videos or reading materials, the platform emphasizes active participation through guided signing tasks, real-time feedback mechanisms, and scenario-based exercises. Users engage with content ranging from alphabet recognition to short conversational interactions, thereby developing both ASL comprehension and sign production abilities.

Gamification elements such as progression tracking, a leaderboard system, and single-player and multiplayer game-based activities. These elements are included to maintain

engagement and encourage repeated practice. Additionally, a computer-vision-based component enables real-time sign recognition, providing users with immediate feedback during practice sessions. Together, these features create a system that makes ASL learning more interactive and focused on practice.

Learning Outcomes

The system is designed to support measurable improvements in both recognition and the ability to make AS signs. Learning is evaluated across three main criteria. Firstly, learners are expected to demonstrate improved retention of the core vocabulary, most of the foundational set of signs introduced in the early units. Second, users should be able to construct simple sign sequences with increasing accuracy, moving from isolated signs to completed expressions. Finally, learners are expected to demonstrate improved confidence and usability when engaging with ASL in real-world contexts. These outcomes reflect both cognitive understanding and motor skill development, aligning with the combined nature of ASL learning.

Target Audience

SignQuest is intended for beginners with little or no prior exposure to American Sign Language. This includes students, educators, and individuals who may encounter ASL in social, academic, or professional settings and want to develop basic communicative skills or at least the ability to recognize signs. The platform also supports secondary users, such as early-stage ASL learners who want additional practice beyond formal instruction. The system is primarily designed for self-paced learning, making it suitable for users with varying schedules and varying levels of technical experience. Overall, the application targets individuals seeking an easier way to learn ASL, focusing on real-world communication rather than formal study.

Background and Related Work

Educational Context and Psychology

The rise in popularity of using technology to learn a language is primarily a result of technology's ability to create accessibility, flexibility, and interactive applications. These digital

platforms enable language learning and maintain engagement outside traditional classrooms. Notable platforms like Duolingo and Babbel have become household tools for language learning. Their success stems not only from its online accessibility but also from its ability to cover the three pillars of Universal Design for Learning (UDL): engagement, representation, and action. A recent study found that users of language-learning applications showed substantial improvement in vocabulary, reading, and listening skills [1]. These results stem from the ability to receive instant feedback and guided, repeated practice, as seen in platforms like Duolingo. They are difficult to implement in conventional classroom instruction without these platforms.

Time spent learning a language is not the only factor that leads to success; the content, methodology, task structure, and level of interactivity are also important, all of which are associated with gamification and correlated with the UDL pillars [1]. Offering a variety of content across different learning objects is important to users' success, as done in SignQuest. A literature review during the proposal and continued through development found that technology-based language learning not only improves language skills but also metacognitive skills, positive attitudes towards learning, language learning retention, and more [2]. These benefits of digital language learning depend on repetition and the mechanisms involved in language learning, leading to higher language proficiency for users.

SignQuest successfully applies UDL principles to ASL by using diverse learning objects, such as real-time sign detection, structured lessons, and a variety of games, all of which provide immediate visual feedback. Our application provided engagement, representation, and action elements in ASL, which historical aspects has been taught through static methods such as copying from a video.

ASL Challenges

Learning ASL posed a unique challenge compared to other languages due to its visual and physical aspects, which became evident during prototype development and planning. There are three functional movements involved in ASL, as well as parameters that aid in phonological awareness (the ability to manipulate the sound structure of languages) [3]. See Figure 1. These

different actions and parameters involve expressing not unique words but also sentence structures, such as asking questions or making statements. Grammar also needs to take into account, and there are three important concepts to understand:

1. Morphology: how the words are grammatically formed [3].
2. Semantics: the meaning of the words and context [3].
3. Syntax: the rules of forming sentence structure [3].

After exploring available data sets that express these parameters, specifically morphological and semantic expressions of words, we found in the ASL-LEX 2.0 data set [4]. However, it is important to note that, as with other spoken languages, ASL has regional and sociolinguistic accents and dialects [5]. This made creating an ASL application extremely difficult, as sociological and regional biases might not be accounted for due to the significant resources required to implement all accents and dialects fully, or to find a dataset without any biases. However, the ASL-LEX 2.0 dataset is standardized, without dialects or accents [4]. This standardized dataset is spoken by a single individual, providing a consistent and dialect-neutral baseline for the learning objectives. Furthermore, parameters and functional movements can be broken into different linguistic categories of producing signs. These categories usually involve different movements for different grammatical implementations, shortcuts or spelling. The following are the main linguistic categories:

1. Fingerspelling: spelling words with the alphabetical hand signs [3, 5].
2. Lexical Signs: a fixed sign for one word [3, 5].
3. Classifiers: hand shapes representing nouns [5].
4. Non-manual markers: provide context to the sign through action [5].

Given the limited use of trainable datasets for classifiers and non-manual marks, and the constraints of real-time computer vision detection of ASL (specifically motion), the translation

feature had to be scaled down to focus on fingerspelling. Of course, there is potential work to implement lexical signs, classifiers, and non manual markers; however, purchasing potential datasets must be considered in this case. As for other learning objectives, such as dictionaries, games, or lessons, SignQuest utilizes ASL-LEX 2.0. ASL-LEX 2.0 uses all of the main linguistic categories; however, primarily lexical signs and distinct hand movements [4]. By eliminating an extensive implementation of fully body non-manual markers (only those near the face and requiring minor movements) or complex classifiers, we were able to create a beginner-friendly application that doesn't overwhelm users but still provides a high level of technical functionality, while also eliminating potential technical overhead. Thus, reaching the goals set in the proposal, "A focus on fingerspelling, lexical signs with hand movements and syntax will be implemented. At the same time, morphology and semantics will be briefly covered, without implementing full-body and face detection. These different functional movements and concepts can be covered conceptually, but are difficult to implement in an ASL detection model".

Gamification

Gamification is the integration of game design principles, such as rewards, leaderboards, badges, challenges, and more, into applications to increase engagement, retention, and academic performance [6]. However, the use of gamification in language learning technologies and its effectiveness differ from its use in other educational subjects. By understanding ASL's learning objectives, SignQuest can implement Gamification without being overly invasive. These learning objectives prioritize vocabulary, sentence structure, and grammar, with less popular objectives involving pronunciation and reading comprehension [7]. SignQuest leverages gamification features to achieve these learning objectives. Empirical studies have shown that gamification directly correlates with increased engagement, including behavioural, emotional, and cognitive engagement, all of which are crucial for learning skills that require extensive practice, such as language learning [7]. Thus, implementing gamification helps achieve the ASL learning objectives of our application.

An important consideration we had when implementing gamification in SignQuest was

the potential for demotivation [7]. We considered that users might get discouraged or overwhelmed by competitive features like leaderboards or badges if they are disappointed with their results. On the other hand, if the results of leaderboards, badges, or other gamification features are related to a user's struggle in a particular field, these features can be used as learning analytics to indicate which fields they need improvement in [6]. Ultimately, SignQuest uses competitive features of gamification, isolated in-game-based learning objectives. In contrast, lessons use achievements and progress bars, aiding progress tracking and encouraging confidence and repetition, all of which are crucial when developing language skills.

Technical Foundation

SignQuest utilizes interactivity and diverse learning objects, by leveraging real-time computer vision techniques to detect ASL signs. In terms of the ASL detection model, the tools in Figure 2, were attempted to be used in the first version of the model, based on the paper "Real-Time American Sign Language Interpretation Using Deep Learning and Keypoint Tracking" [8], however after iterations of training and modifying models there are some slight differences. The paper outlines a potential pipeline for a model as such: The process begins with OpenCV accessing the camera. Then the MediaPipe BlazePalm detector finds the hands in each frame. Next, YOLO (You Only Look Once) with the use of MediaPipe detects and classifies 21 keypoints, a model commonly used for real-time object detection [8].

However the architecture of this pipeline causes difficulty in server communication due to large model weights and library design of MediaPipe. The model in SignQuest utilizes YOLOv11, a computer vision model focused on real-time detection and classification [9]. This model has an increase in performance, speed and efficiency of design compared to its predecessors, YOLOv8, YOLOv9 and YOLOv10 [9]. It was trained on many datasets but ultimately we used ASL-LEX 2.0 dataset [4]. Implementation of the models architecture and server communication is discussed in Technical Implementation.

App Design

The application was designed with Smart Learning Objectives in mind to create an autonomy-supportive learning environment. By providing authentic tools, learners are given the opportunity to determine their own learning process and learn from mistakes within the learning environment. The design balances the four lenses of Educational UX: desirability, feasibility, viability, and learner efficacy.

Learning Objects and Curriculum

Lessons are conducted using Reusable Learning Objects (RLOs). RLOs serve as "LEGO blocks" and are used to design experiences tailored to learners' needs. Lessons are microlearning-based and structured for maximum efficiency, with 5- to 10-minute lesson plans to optimize learner efficacy. Quiz questions incorporate retrieval practices, which are considered one of the most effective teaching methods, favouring active recall over passive rereading [2]. The dictionary serves as a comprehensive RLO library, supporting learner autonomy by allowing users to choose their own study paths. To minimize cognitive load, the interface utilizes concise labels, a consistent grid, and a clear search/filter hierarchy. We also implemented game-based learning to foster intrinsic motivation through visible progress and immediate feedback. Sign Memory targets visual recognition and receptive fluency. Sign Battle is a multiplayer challenge that introduces time pressure to strengthen retrieval speed. Sign Combo is a simulation requiring users to assemble sequences, reinforcing syntactic structure. Translator is an authentic tool that provides multiple means of representation. By converting user-signed input into text, it offers faster feedback and encourages experimental learning.

Core Features

We incorporate multiplayer architecture that uses competition through leaderboards and games to build community and increase engagement. Our application tracks detailed analytics on performance metrics to ensure challenges remain achievable and matched to the learner's skill level. Also, our application uses interleaving by mixing different topics to force a productive struggle. We have adaptive practice sessions that use spaced repetition to strategically reintroduce

signs and help prevent forgetting. Finally, we use real-world simulations that provide immersion and narrative-driven stories, allowing users to experience real-world conversations without the consequences.

Universal Design for Learning

In line with the UDL framework, the application is designed to support learners with diverse abilities. To provide multiple means of representation, the application delivers content through high-definition videos, visuals, and text. We also have many forms of engagement through gamification, progressive difficulty with lessons unlocking in sequence to prevent overwhelming users. For our questions we use a two-attempt system to give users a second chance at questions. Audio reinforcement is also used with audio cues that encourage correct answers and identify incorrect responses through different tones. We also offer multiple ways to act and express, including clicking/tapping the answer, entering text, and drag-and-drop questions. Finally, we implemented many accessibility features that can be turned on or off, including high-contrast dark mode, audio, and dyslexia-friendly font. We followed UI best practices to make it easy for users to navigate the application and find instructions for completing a task.

Technical Implementation

Lessons are structured to be unique each time by randomly selecting three to four signs to teach, then immediately testing them across multiple question formats. If a question is answered incorrectly, it is re-queued at the end of the session, so the learner must answer it correctly before finishing. The accuracy system uses a Leither Box algorithm to sort signs into tiers, automating the spaced repetition. We also have a sign sequencer engine, its purpose is to create sentences by stitching individual videos together. The SignQuest application is hosted using Microsoft Azure, where the frontend is deployed as a Static Web App (built on Next.js) and the backend runs as an App Service container using FastAPI [10]. The frontend handles the user interface and communicates with the backend through API requests. The backend processes these requests (such as sign detection) and returns results, allowing the system to function as a full client-server architecture. Specifically, when a user uses the translator feature, the frontend captures webcam

frames at 100ms intervals and sends them to the FastAPI backend as a base64-encoded JPEG. See Figure 3 for Server communication. The backend server then uses OpenCV to decode each image using the YOLOv11 model to perform hand detection and letter classification in a single instance. Note that the YOLOv11 model was trained on “American Sign Language Letters Dataset”, a 27-class ASL alphabet dataset from Roboflow [11]. See Figure 4 for the model pipeline. Users must hold a sign for 0.8 seconds before it is confirmed and appended, thereby reducing accidental inputs. With our server architecture, we can run a lightweight frontend. At the same time, the backend can be scaled on Azure Container Apps independently of the frontend and in line with model demands.

Evaluation Plan

The success of an education application is defined not only by its technical functionality but also by its learner efficacy, which is the measurable improvement in a student’s knowledge and retention.

Evaluation Objectives

The primary objective of the evaluation plan is to determine if the application successfully transitions from a functional prototype to an effective educational tool that achieves its SMART learning objectives. We aim to establish learner efficacy by seeing if the inclusion of desirable difficulties, such as interleaving and spaced practice, improves long-term retention rather than just providing a "fast and easy" user experience. The evaluation aims to assess the desirability and feasibility of the interface, ensuring that the microlearning sessions are engaging enough to maintain consistent user participation without causing cognitive overload. Ultimately, the goal is to confirm that users can demonstrate both receptive fluency and syntactic mastery by correctly assembling sentences in an autonomy-supportive environment.

Participants

For SignQuest, we used a sample size of 15-20 people with limited to no prior ASL experience. The majority of our testers were students from our Educational App Design class, but we also had testers from outside the class, including a current elementary school teacher, a

university student and a high school student. The 3 participants outside of the class used the app while explaining their thought process to catch confusing text, navigation hurdles and more.

Metrics for Success

Success will be measured using quantitative data and analytics to track user performance. The most important metric is user accuracy, which measures how well a user knows an ASL sign. This data is displayed on the home page as an average understanding of all ASL signs. The accuracy metric is also shown in the dictionary page for each ASL sign which gives the user information on what signs they need to improve on. We also have metrics in our game page. We track high scores for each game so they can see if they are improving over time. Our clearest metric of success is the achievement system. We have achievements that cover all aspects of learning, accuracy, learning progression, learning consistency (using the app every day), and high scores. This is a clear way for a user to measure their progress in their ASL journey.

Peer and Feedback testing/Intergration

From our peer feedback, we made many changes. It was mentioned that many screens had unnecessary scrolling. Based on this feedback, we widened the UI and adjusted the layouts across multiple screens to eliminate unnecessary scrolling. Another UI suggestion was to add coloured borders to the sign combo game to make it more like Wordle. These changes were made to improve the instant feedback. A big suggestion was to make lessons dynamic so they can be repeated without seeing the same questions in the same order. This was a good suggestion, which we implemented by randomizing the order of the questions. It was also mentioned that clearer learning outcomes should be set before starting a lesson. We now have a screen with the learning outcomes before the lesson begins. For lessons, it was also suggested to give the user a second attempt instead of immediately telling them they are wrong. This was an easy addition to the app to promote learning and not just giving them the answer. With our prototype, we used a .bat file, which does not run natively on Mac. This was easily corrected by making our application accessible through a link, so it is easy for everyone to access.

Conclusion

Future Work

Although the SignQuest application provides an engaging, dynamic environment for beginners learning American Sign Language, there are some improvements that could be made in the future. One possible improvement concerns the real-time translation model and its future expansion into other applications. Currently, the SignQuest model focuses on fingerspelling due to limited data and the complexity of real-time sign translation. In the future, the model can be trained to recognize entire lexemes and translate them between languages.

The future development of the system should include integrating additional elements of ASL, specifically classifiers and non-manual signs. Non-manual signs, such as facial expressions and head and body movements, play a crucial role in conveying meaning in ASL. The incorporation of these elements into the system will enhance its performance; however, it will require advanced computer vision systems capable of processing full-body data.

Regarding the educational aspect, another possibility is to expand the curriculum to include many more lessons focused on ASL grammar. In addition to the basics currently taught, future versions could provide more instruction on grammar and on creating well-structured sentences with it. In this way, the user could learn to incorporate it into more practical applications and become fluent with it.

Another area for improvement is the program's ability to provide technical feedback to the user while the user uses the translator. The present software can detect and identify signs performed by the user, but future iterations can improve by providing feedback on the accuracy of those signs.

Lastly, personalization capabilities can further enhance this app. The implementation of adaptive learning systems can enable this app to determine each individual's strengths and weaknesses, thereby customizing lessons to their level. The app can automatically detect common signs that a learner finds challenging and then reintroduce them for extra practice. It can also adjust the difficulty of practice questions by gauging whether learners find the task too easy or too

difficult. With these improvements, future versions of SignQuest could become a comprehensive ASL learning platform.

Summary

In summary, this report has explained the design, implementation and evaluation of SignQuest, a web application designed to facilitate learning American Sign Language (ASL). The project was created in response to problems with current approaches to teaching ASL, such as a lack of interactive, feedback-driven practice with signs. With structured lessons, game-based practice, a dictionary and a real-time translator, SignQuest offers learners a comprehensive approach to acquiring ASL skills.

The application was developed using a stack of technologies, with Next.js for the frontend and FastAPI for the backend. A technical innovation used by SignQuest is the YOLOv11 algorithm for real-time sign detection and feedback during a user practice session. Furthermore, several other methods have been used, including gamification, spaced repetition, and varied techniques.

SignQuest demonstrates that modern technologies can provide an engaging language-learning experience outside conventional educational settings. Overall, the application aligns with Universal Design for Learning principles and supports the development of language skills through interactive, regular practice.

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Appendix

Functional Movements	Parameter	How it works
Hand Movements	Handshape	Shape of the Hand
	Movement	The action or path the hand takes.
	Location	Where the sign is made with respect to your body and air.
	Palm Orientation	The orientation of the palm when the sign is made
Body & Face	Non-Manual Markers	Facial grammar, as well as head and body shifts, made with the sign

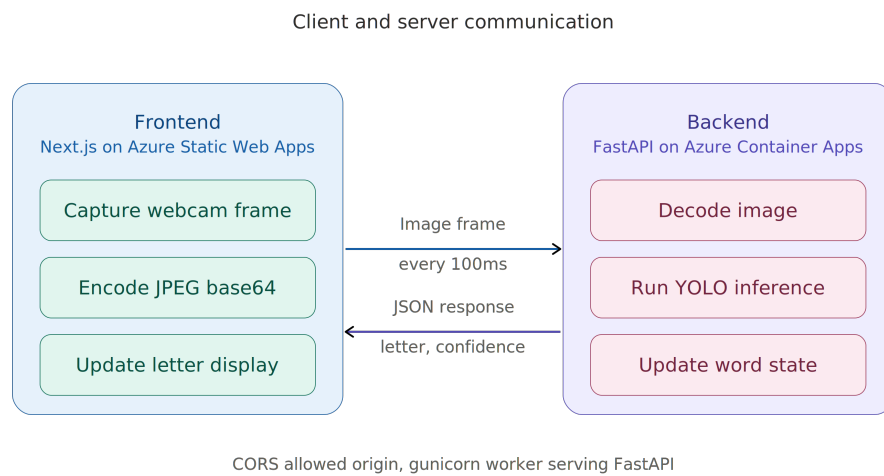
Figure 1

The American Sign Language functional movements and their parameters [3].

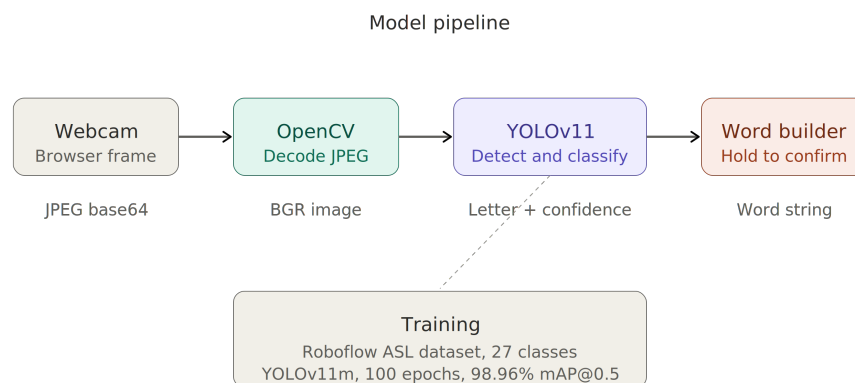
Tool	Purpose	How it works
OpenCV	Camera	Access to the camera
MediaPipe	Detect and tracking	Detects and tracks Hands
BlazePalm Detector	Detect Hand	Helps detect palm and bounding box
Hand Landmark Model	Feature extraction	With the YOLO model, this model is a pretrained CNN with 21 detected keypoints.
YOLO Model	Detect ASL	Classified ASL letters and words using landmarks

Figure 2

Original pipeline from the translator mode from the paper "Real-Time American Sign Language Interpretation Using Deep Learning and Keypoint Tracking" [8].

**Figure 3**

Frontend server and backend server communication for the translator model.

**Figure 4**

Translator model pipeline and training results.